STUDENTS' EXPERIENCES WITH GAMES AND THEIR ACADEMIC POTENTIAL: AN INTERPRETATIVE PHENOMENOLOGICAL ANALYSIS

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Abstract

The purpose of this research project was to explore students' gaming experiences in relationship to engagement and learning to expand our understanding of the essence of such experiences and produce valuable results for educators seeking to create a more student-centered learning environment, facilitate active learning, and increase students' engagement. Utilizing flow theory as theoretical framework and Interpretative Phenomenological analysis, this study explored the gaming and academic experiences of five college students. Participants' interviews, transcriptions, annotations and preliminary analysis led to the emergence of five core themes: (1) manifestations of flow; (2) engagement, expectations, and feelings; (3) health related issues in games; (4) social aspects of games and learning; (5) games, learning, and the real world. The analysis of participants' interviews and additional data pointed out a variety of engaging factors the typical classroom experience lacks, but that were found to be abundantly present in certain games. Several important implications were found and discussed. These included implications related to: engagement and active learning; motivation and feelings; social dynamics; teaching, learning, and the teachers' role; teaching, learning, and the real world; game based learning; institutional support and faculty development; and the gaming field. This study's findings supported the idea of incorporating games and gamification in the college classroom. Transporting certain factors present in games to the classroom in the form of positive and memorable experiences was not only desired by students but also important for their engagement and academic success.

Keywords: engagement, game based learning, flow

Dedication

I dedicate my dissertation work to my family and spiritual mentors. A special feeling of gratitude to my loving wife Mei, my mother, Ana Maria, and my father, Vittorio whose example, help, and encouragement made all possible.

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Chapter 1: Introduction

It is early in the morning and students are seated in their college classrooms for the first time. Soon after introductions, the syllabus is reviewed and the professor starts talking about the different topics that need to be covered for the class. Besides the occasional question or brief discussion, the traditional lecture has started and will continue mostly unchanged in every session, something that drives many students' minds to wonder while the professor's voice fades in the background of their thoughts as they reach for anything available that would make their time in class more enjoyable. Over time, these type of classes become a monotone routine that fail to gain students' attention for more than just a few minutes (Khan, 2012; Robinson & Kakela, 2006; Wilson & Korn, 2007; Young, Robinson, & Alberts, 2009) and lead them to passively receive information in order to robotically complete assignments. Eventually, students become disengaged and their original enthusiasm and excitement about college fades away together with the possibility of becoming actively engaged in their learning. . It is easy to see how in this scenario how students' academic performance would significantly decrease while dropouts would increase, and eventually damage the quality of our educational system.

Students born after 1990, also referred to as generation Z (Geck, 2007), entered a society where technology had progressed significantly, computing devices became pervasive, and information became extremely accessible while instantly available. This generation of students is considered more social, self-directed, and able to quickly process information (Igel & Urquhart, 2012), but it is also more easily distracted (Ding, Guan, & Yu, 2017) and in need of novel ways to grab and retain their attention . While students' ways of learning are evolving, our educational system is confronted with important challenges that demand a departure from conventional teaching methods.

The Research Problem

The effectiveness and engagement potential of traditional teacher-centric lectures have been questioned by many scholars (Aljezawi & Albashtawy, 2015; Frederick, 1986; Khan, 2012; Middendorf & Kalish, 1996; Robinson & Kakela, 2006; Svinicki & McKeachie, 2006; Wilson & Korn, 2007). In parallel, the quality of our educational system continues to fall in comparison to other countries (OECD, 2012), students' grades keep lowering (Payne, 2008), and engagement consistently decreases as students get older (Brenneman, 2016, March 22). While it is generally accepted that students' attention and engagement play an important role in the effectiveness of learning and could be determinant for academic success (Coates, 2005, 2010; Floyd, Harrington, & Santiago, 2009; Jagannathan & Blair, 2013; Krause & Coates, 2008; Trowler, 2010), the use of orthodox teacher-centric lectures still remain the most commonly used teaching method in higher education (Khan, 2012).

In spite of the general agreement among scholars about the benefits of student-centered lessons in comparison to traditional lectures (Granger et al., 2012; Middendorf & Kalish, 1996; Moura & Van Hattum-Janssen, 2011; Robinson & Kakela, 2006; Sahin & Abichandani, 2013), there is still a lack of general best practices and methods for their implementation. Two approaches that have recently gained attention and that are the focus of this research are Game Based Learning (GBL) and Gamification. GBL involves using games for educational purposes while Gamification aims to incorporate game elements such as points, levels, badges, and awards to educational lessons (Deterding, Dixon, Khaled, & Nacke, 2011; Isaacs, 2015; Kingsley & Grabner-Hagen, 2015; "What is Gamification?," 2015).

The increasing number of studies suggesting the academic potential of games (A. Huang & Levinson, 2012; Mayo, 2009; Neef, Perrin, Haberlin, & Rodrigues, 2011; Rosser et al., 2007), suggest that their importance and relevance should not be overlooked. Considering $14.80 billion was spent in the United States on video game content in 2012 (Riley, 2013) and 97% of all American teenagers play video games (Amanda Lenhart et al., 2008), the increasing popularity of these tools present significant academic opportunities. It has been found that when applied correctly, these novel methodologies have the potential to greatly increase students' engagement, attention span, and other related issues that can significantly influence students' academic success (Aljezawi & Albashtawy, 2015; Claman, 2015; Entertainment Software Association, 2014; A. Huang & Levinson, 2012; M. Jong, J. Shang, F. L. Lee, & J. Lee, 2008; Sanchez Martin, Yu-Ju, & Tsun-Ju, 2014; Simões, Redondo, & Vilas, 2013; Tews, Jackson, Ramsay, & Michel, 2015).

Deficiencies in the Evidence

Although some studies have addressed the use of GBL and Gamification in adult teaching and learning (Aljezawi & Albashtawy, 2015; Coleman, 2012; A. Huang & Levinson, 2012; Korolov, 2013; Ryoo, Techatassanasoontorn, Lee, & Lothian, 2011; Tews et al., 2015), the majority of the research have remained confined within the elementary, middle, and high school educational levels. In addition, their application in post-secondary education together with their impact on higher learning have remained largely unexplored (Perrotta, Featherstone, Aston, & Houghton, 2013) and in clear need of further study. By examining college students' experiences with games and Gamification a better understanding can be reached about factors that can make them effective academic tools. Furthermore, understanding how these student-centered methods can influence college students' engagement not only help to improve the quality of teaching at the college level, but also bring us closer to a general set of best practices that will broaden their implementation and integration with more traditional teaching methods.

Significance of Research Problem

The predominant lecturing method in our classrooms (Khan, 2012) is in clear need of variations and alternatives in order to improve our teaching effectiveness. The fact that younger generations are growing up spending an average of 7:38 hours per day involved with various media (Rideout, Foehr, & Roberts, 2010) and the consistent growth of game usage and popularity (Association, 2016; Markert, 2016, May 12), points towards the special relevance, influence, and importance of games in our current generation of students. In higher education, the use of games has been mostly relegated while play is generally regarded as a purely entertainment activity without seriously reflecting on its teaching and learning potential.

Considering the fact that GBL and Gamification can significantly increase students' engagement and have a positive impact in the effectiveness of learning (Entertainment Software Association, 2014; Mayo, 2009; Middendorf & Kalish, 1996; Neef et al., 2011; Robinson & Kakela, 2006; Tews et al., 2015), dismissing games' potential in higher education is a negligent mistake that should not be ignored. Understanding more about students' experiences with games and their engaging qualities help us maximize their academic impact and reach a more common set of practices that could be applied to increase their effectiveness, facilitate their application, and widen their usage in education. Furthermore, considering the current problems affecting our educational system (Brenneman, 2016, March 22; OECD, 2012; Payne, 2008), finding new and more entertaining teaching methods that can gain students' attention for longer periods and place them in the center of their learning is of great importance and determinant in the quest for effective teaching.

Research Question

Qualitative research is interested in "understanding the meaning people have constructed, that is, how people make sense of their world and the experiences they have in the world" (Merriam, 2009, p. 13). Similarly, qualitative studies using an Interpretative Phenomenological Analysis (IPA) approach focus on questions that are open, and ask about "people's understandings, experiences and sense-making activities" (Smith, Flowers, & Larkin, 2009, p. 47). Smith et al. (2009) remark how the words "exploring", "investigating", "examining", and "eliciting", are most commonly used to state IPA researchers' goals and actions while Creswell (2012) indicates how qualitative studies should use the words "what" or "how" in order to "explore a central phenomenon" (p. 138) .

This study proposed the following overreaching research question: what are students' gaming experiences in relationship to engagement and learning? Finding answers to this research question allowed this study to explore the gaming "phenomenon", or "experience", and expanded the understanding of its essence. In turn, this produced valuable results for educators seeking to create a more student-centered learning environment, facilitate active learning, and increase students' engagement.

Positionality Statement

Being a student and a teacher for many years and in different countries has given me an interesting perspective on the problem of students' engagement in the classroom. I have experienced firsthand the difference between teaching styles and methods and how some traditional lectures can lose students' attention in just a few minutes while others that are mixed with interactive elements can create more engaging learning environments. In my experience, the instructional methods that a teacher uses in class can make a significant difference in students' engagement and is something that should be studied in detail in order to promote students' academic success (Coates, 2005, 2010; Floyd et al., 2009; Jagannathan & Blair, 2013;

Krause & Coates, 2008; Trowler, 2010).

I believe that being in the classroom should not only be about transmitting information, but also about sharing ideas, creating relationships, motivating students, encouraging critical thinking, having fun, and generating memorable experiences. As a scholar-practitioner, my positionality has giving me a unique perspective in the problem of students' lack of engagement. It has also lead me to research new teaching methodologies like GBL and Gamification, that may address some of our current students' engagement problems and enrich their learning experiences.

When thinking about games as tools to promote academic engagement, I realize that my personal experiences and beliefs are influenced by specific factors that are part of my positionality and can influence my research (Jupp & Slattery, 2010). One of these factors is my relationship with games. I will never forget when I was about eight years old and received as a Christmas gift an "Atari 2600" electronic game console complete with several games and controls. I remember how the games for this console were accompanied by manuals and instructions that would give clues to solve different puzzles and quests. These manuals were all written in English, a language that was strange to me at the time; and that represented a much greater challenge to overcome in my determination to reach the games' objectives. These challenges drove me to learn how to solve problems using a trial and error approach, learn how to use my first dictionary, develop an avid interest in literature, and take my first steps towards learning a new language.

My interest in games continued in high school with the introduction of personal computers. Computers not only increased the sophistication and interactivity of games, but introduced me to much more complex challenges that required specific knowledge and expertise (e.g., installing and troubleshooting new hardware and software). Furthermore, in the South American city where I grew up, the information that I needed was not easily available, accessible, or comprehensible for me in those early years. Yet the challenges presented by some games motivated me to visit bookstores and libraries in what I considered a quest for knowledge in order to help me solve a technical problem or a game's riddle. This quest also drove me to socialize with classmates who were playing the same games or trying to solve similar problems. I have many memorable moments where my friends and I would brainstorm ideas and share clues that would allow us to overcome a technical obstacle or game's quest. Games became a stimulating and shaping mental challenge in my life; a catalyst for study, research, collaboration; and, in many ways, an initiation to my engineering "problem solving" training.

These personal experiences together with my professional background will bring unique advantages to my selected research process. For example, my engineering training combined with my academic knowledge, technical expertise, and teaching experience, prove invaluable in meeting the complexities associated with implementing GBL and Gamification strategies in the classroom. The results obtained from this research will better inform my teaching practices and improve the quality of my work as an educator and scholar-practitioner.

Researcher's Influence and Transparency. My experiences and identity have inspired me toward choosing my proposed line of research. Although it is my belief that carefully planned and organized mixing of what is generally labeled as "fun" with "work" bring significant benefits to students, I realize that my definitions of these terms are influenced by personal factors including my positionality (Jupp & Slattery, 2010). Having been born in Venezuela from an Italian family that later immigrated to the United States of America, it is obvious that positionality in relationship to my cultural frameworks, as defined by Jupp and

Slattery, could be a key bias factor that requires reflection in order to understand all of its dimensions and ramifications (Briscoe, 2005). Briscoe (2005) argues that “it seems incontrovertible that differences do matter in the way that we interpret our social world” (p. 35), inferring that identifying personal influencing traits and experiences becomes necessary and vital for any researcher. Being aware of these differences and placing careful attention on avoiding “deficit thinking” (Jupp & Slattery, 2010) or any form of discriminatory profiling is important for any research.

In order to maintain transparency, I was committed to remain aware of my positionality and its influence by having a clear understanding of the meaning of “self” and realizing how we are all interconnected in our society. Briscoe (2005) reflected on how we are united in our differences and how “the *other* is always within us” (p. 30). This brilliant remark illustrates how each of us contain all of humanity´s differences and traits while remarking the significance that this realization brings to our lives as research-practitioners.

While understanding the researchers’ influence in a study is important, maintaining research transparency is “an essential foundation for rule-governed and intersubjectively valid social science research” (Moravcsik, 2014, p. 49). This study’s transparency was ensured by pursuing the highest quality of research. The research’s quality was maintained by providing scholars access to the data collected, used methodology, processes, theories, and all related materials, with the goal of producing strong and trustworthy results that will contribute to the study of students’ engagement and Game Based Learning.

Flow Theory

Anfara and Mertz (2006) defined theoretical frameworks as “an empirical or quasi-empirical theory of social or psychological process at a variety of levels (e.g. grand, mid-range, and explanatory) that can be applied to the understanding of phenomena" (p. xxvii). Theories of motivation and engagement are most applicable to this study. Among the different motivational and engagement constructs, intrinsic motivation principles align best with this research. Motivation is intrinsic when subjects display an innate propensity, interest, and pleasure towards completing a certain kind of activity (Sansone & Harackiewicz, 2000). Games are a well-known source of intrinsic motivation and engagement that if harnessed correctly can be utilized as a significant learning tool (Aljezawi & Albashtawy, 2015; A. Huang & Levinson, 2012; M. Jong, J. Shang, F. L. Lee, et al., 2008; Mayo, 2009; Neef et al., 2011).

The apparent reciprocal relationship between engagement and motivation (Reeve & Lee, 2014; Saeed & Zyngier, 2012) should be considered when studying gamers' experiences. One of the theories that consider this relationship and is appropriate for this study is flow. The concept of "flow", also known as "optimal experience", is a mental state that occurs during certain activities where the level of challenge and the subject's skill level are at their highest (Csikszentmihalyi, 1975). Flow is regarded as a "strong conception of intrinsic motivation" (Moneta & Csikszentmihalyi, 1996, p. 279) that can be used as a lens to examine students' gaming experiences. By utilizing flow as the theoretical framework of this study and guide us towards increasing engagement in the classroom.

Flow's Definition and Context

Flow was introduced by Csikszentmihalyi (1975) and it is used to describe a dynamic state of enjoyment experienced by some people when undergoing activities that meet a specific criteria. In a flow state, subjects notice how "action follows upon action according to an internal logic that seems to need no conscious intervention by the actor. He experiences it as a unified flowing from one moment to the next" (Csikszentmihalyi, 1975, p. 36).

Flow is experienced in certain activities where clear goals are defined and when “tasks are within one’s ability to perform” (Csikszentmihalyi, 1975, p. 39). Csikszentmihalyi also described flow as an “optimal experience” that offers just the right balance between challenge and skills. This balance is of crucial importance for flow to occur and is a key tenet of the theory.

Moments of flow fall into a limited area or “zone” of experience (see Figure G1). Csikszentmihalyi describes this flow zone stating,

> If challenges are too high, one gets frustrated, then worried, and eventually anxious. If challenges are too low relative to one's skills, one gets relaxed, then bored. If both challenges and skills are perceived to be low, one gets to feel apathetic. But when high challenges are matched with high skills, then the deep involvement that sets flow apart from ordinary life is likely to occur. (Csikszentmihalyi, 1997, p. 30).

In other words, the original flow model presents a series of fluctuating states that subjects can experience during an activity depending on their level of challenge and skill. These states conform the foundation of what has been referred to as the “four channels model of flow” and fall into four main categories: apathy, boredom, anxiety, and flow. In these optimal flow experiences, goals are clearly defined while feedback is immediately provided. Although flow is closely related to enjoyable activities, it goes beyond feelings of well-being and includes characteristics that can be associated with deep immersion, merging of action and awareness, deep focus and attention, lack of self-consciousness, and “not requiring goals or rewards external to itself” (Csikszentmihalyi, 1975, p. 47).

People experiencing flow are completely immersed in the task at hand and their attention is focused on the goal in such a way that their awareness is merged with action while their sense of self is lost in an environment where "there is no need to negotiate rules" (Csikszentmihalyi, 1975, p. 73). A flow state produces in subjects a sense of control without causing worry about losing control, while simultaneously being intrinsically motivating and requiring a deep sense of attention that can distort users' sense of time and location. Flow's relationship with the study of optimal experiences is considered to be part of the Positive Psychology field in charge of the "scientific study of ordinary human strengths and virtues" (Sheldon & King, 2001, p. 216) and their valued subjective experiences of "well-being, contentment, and satisfaction (in the past); hope and optimism (for the future); and flow and happiness (in the present)" (Seligman & Csikszentmihalyi, 2000, p. 5).

Although Csikszentmihalyi's concepts may have been influenced by Abraham Maslow's (1962) studies on "peak" experiences, flow experiences are unique in that they are more frequently experienced and are far from being rare or extreme (Rodríguez Sánchez, 2009). The original concept of flow was later revised by Fausto Massimini and Carli (1988) who included four additional states to the model: relaxation, arousal, worry, and control (see Figure G2). This expanded version of flow was embraced by Csikszentmihalyi (1997) and referred to as "the eight channels flow model" where moderate skills and challenges play a more important role generating additional states or feelings: moderate challenge paired with low skill generate worry, moderate challenge paired with high skill generate feelings of control, moderate skills paired with low challenge generate relaxation, and moderate skill paired with high challenge generate arousal.

Csikszentmihalyi's original study of flow was joined by many other scholars that looked to expand and apply the theory to a diverse set of fields. For example, Csikszentmihalyi and Lefevre (1989) researched the presence of flow in a work environment and compared it to its occurrence in leisure activities looking to answer whether "the quality of experience was more influenced by whether a person was at work or at leisure or more influenced by whether a person was in flow" (p. 815). Similarly, F. Massimini, Csikszentmihalyi, and Carli (1987) aimed to confirm the importance of challenge and skill in flow. They found that the ratio between challenge and skill was a fundamental predictor of optimal experiences, boredom, and anxiety.

Flow's Connections with Games and Learning

Flow theory is considered innately related to games and learning. In fact, Csikszentmihalyi was interested in the study of happiness since his childhood in wartime Europe, where he spent some time in a prison camp and developed a fondness for chess. Chess helped him to overcome the difficult conditions of his imprisonment and became one of his first experiences with flow (Sobel, 1995).

The inherent link between games and the flow state was later emphasized by Csikszentmihalyi (1975) when he stated that games are "obvious flow activities" while "play is the flow experience par excellence" (pp. 36-37). Since not all games are made equal and a game one person finds fun may be disliked by another person, Csikszentmihalyi clarified that not all games or activities guarantee flow because experiences are individual, personalized, and subjective. The connection between flow and games is not only evident but also reflected in many recent studies that have utilized flow as a lens to frame their research on game design, virtual worlds, and learning (Chen, 2007; Cooper, 2009; Y.-C. Huang, Backman, & Backman, 2010; Kiili, de Freitas, Arnab, & Lainema, 2012; Kiili, Lainema, de Freitas, & Arnab, 2014; van Schaik, Martin, & Vallance, 2012; Winberg & Hedman, 2008).

Csikszentmihalyi emphasized the close relationship between flow and learning, stating that flow "acts as a magnet for learning" (Csikszentmihalyi, 1997, p. 33). He explained how, in the "arousal" state, the person can return to flow naturally by acquiring new skills while learning is also incentivized when a subject increases the level of challenge in an activity. Following this line of reasoning, learning can be considered an integral part of flow since the conditions and factors that lead to it require subjects to self-improve in order to match their skills to the emerging challenges. This cyclical process is very similar to the one utilized in many games and frequently used in GBL.

Flow's Methods and Relevancy to this Study

Although flow subjective experiences may be perceived to be naturally aligned to qualitative methods, a large portion of flow studies focus on quantitative methods that measure the occurrence of flow in different circumstances (Kefor, 2015). When trying to measure the frequency of flow experiences, one of the most commonly used tools in quantitative studies is the Experiential Sampling Method (ESM) where respondents are sent electronic signals during an extended period of time in order to instruct them to self-report their flow experiences as they are happening (Csikszentmihalyi & Larson, 1987; Csikszentmihalyi, Larson, & Prescott, 1977). ESM has become useful and important in certain quantitative studies, but it has a number of challenges and disadvantages that can limit its application. For example, since subjects are normally asked to stop their activities to fill out reports on their experiences, this method can become intrusive and even disruptive (Zirkel, Garcia, & Murphy, 2015), causing the interruption of deep flow, something that would "destroy the phenomenon" (Nakamura & Csikszentmihalyi, 2014, p. 101). In addition, due to the high frequency of the signals sent, ESM studies can expose participants to time consuming and extended periods of study requiring a "high level of commitment from the participants" (p. 7) and, in turn, increasing the frequency of drop-outs from a study.

While ESM can be very useful in quantitative studies that aim to measure the frequency and intensity of flow experiences, other methods are needed to understand flow and its essence. For example, in-depth interviews can be used to provide rich insights on past experiences and are indispensable especially when direct observation of a studied phenomenon, behavior, or condition is not possible (Merriam, 1991). Flow questionnaires can be used in order to propose "definitions of flow and asks respondents to recognize them, describe the situations and activities in which they experience flow, and rate their subjective experience when they are engaged in flow-conducive activities" (Moneta, 2012, pp. 24-25). These methods more closely align with the nature of this study, as they are expected to provide a deeper view of students' flow experiences with games. The subjectivity and individual nature of flow experiences in games together with the oversaturated use of ESM as a primary data collecting tool in flow studies (Kefor, 2015), calls for more qualitative approaches to studying flow theory that can offer different perspectives and insights on the gaming phenomena and its academic implications.

Some critics have labeled flow as a western psychic phenomenon (Sun, 1987) that borders with the ethereal or even the mystical. These claims are based on the fact that whether one is in flow or not "depends entirely on one's perception of what the challenges and skills are" (Csikszentmihalyi, 1975, p. 51). Although flow experiences are obviously subjective and may vary from culture to culture (Csikszentmihalyi, 1975), flow's dynamic process remains unchanged regardless of the person experiencing it.

Flow has consolidated itself in the field of Positive Psychology as an important theory. Although the importance of the relationship between challenges and skills have been scrutinized by some authors (Løvoll & Vittersø, 2014), substantiated negative critiques are sparse and difficult to find. Flow theory's tenets and details are not only critical in trying to gain deeper